

Regime Viability on the Northern Cod Stock: An Exact Certification of Obstruction Structure on Public Assessment Data

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Abstract

The viability machinery of the obstruction calculus has been developed and validated on deliberately small exact instances. This paper applies it to a real assessment record — the Northern cod stock (NAFO 2J3KL), using the published assessment input series, the 2025 reassessment series, the official removals tables, the long-duration catch reconstruction, the research-vessel survey index, and an independently monitored forage-fish covariate — and certifies, in exact rational arithmetic, the obstruction structure that the historical collapse exhibited. The worst realized net multiplier of the 1983–1989 decade is $\frac{41800}{45147} < 1$: even the reference decade was not robustly viable at the 276 kt auxiliary threshold, whose level the limit reference point (884.6 kt, the decade’s own mean) encodes. Across 1992–1993 the realized net multiplier was $\frac{2021}{7639} \approx 0.265$ while removals fell to 11,392 t: the decline proceeded with the fishery effectively closed, a witnessed obstruction in the dynamics rather than the catch level. The breach-time certificates match the realized timeline exactly (below the auxiliary threshold within two years of the 1991 reading; the reference point already breached in 1991), the research-vessel survey — an instrument independent of the assessment model — falls by a certified factor of $\frac{21797}{2127417}$ over 1989–1994, the covariate collapses by $\frac{138}{5783}$ in 1991, one step before the mortality diagnostic crosses $M = 1$, and the post-moratorium record re-crosses the auxiliary threshold by 2016 while never approaching the reference point. Monitoring, not catch level, is the certifiable instrument. Every quantity is regenerated by an accompanying verification script in exact rational arithmetic.

Keywords: viability; obstruction; regime shift; Northern cod; exact computation; monitoring design; limit reference points

0.1 Introduction

The viability and obstruction machinery of the calculus — survivable sets, robust multipliers, breach-time certificates, and probe design — has so far been developed and validated on deliberately small exact instances. This paper applies it to the Northern cod stock (NAFO divisions 2J3KL) and certifies the obstruction structure of its collapse on the public assessment record. The forecast layer

of the same record — the scoring of competing forecast ladders and the 2022–2023 observation suspension as a review-timing event — is certified in a companion paper (Abaee, 2026c, Section 3.10); the present paper certifies the viability layer. The two are companions by construction: the forecast ladder prices what can be predicted, this paper prices what could have been kept safe.

0.2 Data and provenance

Seven locked public files are used, each cross-verified against its primary source at archival time (Abaee, 2026d): (i) the NCAM 2016 assessment input series (DFO, 2016, Table A2): spawning stock biomass in kilotonnes, 1983–2015, cross-checked 33 of 33 years against the primary report; (ii) the xteNCAM 2025 reassessment SSB series (Regular et al., 2025, Table 17), 1954–2024, with 95% bands and SSB-over- B_{lim} ratios, implying $B_{\text{lim}} = 276$ kt; (iii) the official removals table (Regular et al., 2025, Table 1), 1954–2023; (iv) the catch reconstruction (Schijns et al., 2021, Table 1), whose recent rows agree with (iii); (v) the fall research-vessel survey abundance index (Schijns et al., 2021, Table 3, citing DFO 2021b) — an instrument independent of the assessment models; (vi) the RAM Legacy database per-year extract for this stock (free, 2024 release), 1850–2021, extending the assessment series to 2021; and (vii) the capelin acoustic biomass series (observed years only, no interpolation). Two discipline rules are respected throughout: the mortality and fishing-mortality columns of the assessment input file are diagnostics only (dynamics are never built on them), and every annual multiplier certified below is a *net* realized quantity — it conflates the mortality regime with all removals and is certified as such, not decomposed.

0.3 The certified windows

Proposition 1 (the two windows are sub-unitary at worst case). *From the Table A2 series: the 1983–1989 window’s six annual net multipliers have minimum $\rho_g = 836.00/902.94 = \frac{41800}{45147} \approx 0.9259$ (1986), and the 1991–1993 window’s two annual net multipliers are $\frac{38195}{73451} \approx 0.520$ (1992) and $\frac{2021}{7639} \approx 0.265$ (1993). Both windows’ worst-case multipliers are strictly below unity.*

Proof. Each multiplier is the exact ratio of consecutive Table A2 SSB entries in kt over 100, computed in the script as a rational; the minimum and the two bad-window values are the displayed fractions by direct comparison of the six and two ratios respectively. Each displayed fraction is below 1 because its numerator is smaller than its denominator ($41800 < 45147$, $2021 < 7639$). $\square$

Proposition 2 (the moratorium obstruction). *Across 1992–1993 the realized removals fell from 40,956 t to 11,392 t (official removals table), approaching the closed fishery; the realized net multiplier of the same pair of years is $\frac{2021}{7639} \approx 0.265$. At the realized rates, removing the fishery did not arrest the decline: the obstruction is witnessed in the dynamics, not in the catch level.*

Proof. The removals are the certified table rows for 1992 and 1993; the multiplier is Proposition 1’s 1993 ratio, whose numerator and denominator are the realized 1993 and 1992 SSB values. A multiplier of $\frac{2021}{7639}$ with removals at 11,392 t against a biomass of tens of thousands of kilotonnes certifies that the decline is dominated by the non-fishing component of the net rate; the statement is a witness about the realized window, not a counterfactual claim about other catch paths. $\square$

0.4 Breach-time certificates

Proposition 3 (breach bounds and the realized timeline). *With $F_{\text{lim}} = 276$ kt: (i) from the end-1991 reading 734.51 kt, one step at the window’s worst multiplier already certifies breach, since $\frac{2021}{7639} \leq \frac{27600}{73451}$; (ii) under the realized two-step sequence the stock is below F_{lim} in exactly two years — $101.05 < 276$ in 1993, the xteNCAM file’s SSB-over- B_{lim} ratio 0.29 at 1993 confirming the same fact in the independent assessment; (iii) under the limit reference point = 884.6 kt the 1991 reading 734.51 is already below the reference point. The certificates match the realized timeline exactly.*

Proof. (i) is the exact inequality $\frac{2021}{7639} = \frac{2021 \cdot 73451}{7639 \cdot 73451}$ against $\frac{27600}{73451}$, checked by cross-multiplication of integers in the script ($2021 \cdot 73451 < 7639 \cdot 27600$). (ii) is the direct comparison $101.05 < 276$ on the Table A2 row for 1993, with the xteNCAM ratio column read as the locked decimal 0.29 at 1993. (iii) is $734.51 < 884.6$. No step of the proof uses floating point. $\square$

0.5 The survey instrument and the catch record

Proposition 4 (the model-independent survey confirms the collapse). *The fall research-vessel survey index falls from 2,127,417 (1989) to 1,117,670 (1991), 239,740 (1992), 90,709 (1993), and 21,797 (1994): a certified factor of $\frac{21797}{2127417} \approx 0.0102$ over 1989–1994, and $\frac{239740}{1117670} \approx 0.214$ across 1992 alone. The instrument is independent of the assessment models that produced the biomass series, so the collapse is witnessed twice, by two instruments.*

Proof. All four values are rows of the locked survey file; the factors are their exact ratios. Independence of the assessment model is a provenance property of the file (the survey index is an input collected by research vessels, not

an output of the assessment), recorded at archival time. $\square$

Proposition 5 (the reconstruction certifies a near-closed fishery). *The catch reconstruction gives removals of 1,314 t (1994) and 413 t (1995) — four and five orders of magnitude below the fishery’s operating scale in the preceding decade — and its 2015 row (4,436 t) agrees with the official removals table and with the assessment report, a cross-file consistency check. The Proposition 2 witness therefore stands on removals effectively at zero.*

Proof. The three rows are read from the locked reconstruction file; the agreement at 2015 is the exact equality $4436 = 4436$ across the two files (and matches the assessment report’s removals figure, as recorded at archival time). The comparison with the preceding decade’s scale is the ratio against the 1983–1989 removals rows, all above 130,000 t in the same file. $\square$

0.6 The reference decade was already marginal

Proposition 6 (the reference point is the decade’s own mean, and the decade was not robustly viable). *(i) The limit reference point 884.6 kt equals the 1983–1989 mean of the Table A2 SSB series (884.58, rounded), and the decade’s readings stay within $\pm 6.4\%$ of it: the ratios run from $\frac{4180}{4423} \approx 0.945$ (1986) to $\frac{94075}{88460} \approx 1.064$ (1987). (ii) Under the window’s worst-case multiplier $\rho_g = \frac{41800}{45147}$ sustained from the 1989 reading 921.66 kt, the stock breaches $F_{\text{lim}} = 276$ kt within sixteen years: $\rho_g^{16} \leq \frac{27600}{92166} < \rho_g^{15}$. No removals policy certifies indefinite above-floor viability at worst-case realized multipliers, since even zero removals inherits the net multiplier’s strict decline.*

Proof. (i) The mean is the exact rational sum of the seven readings over seven, equal to $\frac{88458}{100}$ up to the certified rounding of the record; the displayed ratio bounds are the exact minimum and maximum of the seven ratios. (ii) The sixteenth and fifteenth integer powers of $\frac{41800}{45147}$ are compared against $\frac{27600}{92166}$ by exact cross-multiplication in the script. The final sentence: any policy with non-negative removals has a net multiplier at most the zero-removals multiplier, which at worst case is $\rho_g < 1$, and a strict contraction crosses any fixed positive floor in finite time (at most $\lceil \log(F_{\text{lim}}/z_0)/\log \rho_g \rceil$ steps, the integer certified by the exact power comparison). $\square$

Corollary 7 (what the reference point encodes). *The reference point’s construction — the 1980s mean — is precisely the certification that the healthy decade sat at its own failure boundary: the stock was marginal at B_{lim} -scale viability before the regime shift, which is the viability reading of the same fact the assessment literature records as the reference point’s provenance.*

Proof. Immediate from Proposition 6(i)–(ii): the mean-rule fixes the boundary inside the decade’s realized range, and the worst-case contraction certifies non-viability of the sustained worst case within sixteen years. $\square$

Table 1: The certified windows and instruments (all quantities exact; SSB in kt, removals in t, survey index in model-independent units).

Quantity	Value	Source file
Decade worst mult. (1986)	41800/45147	Table A2
1992, 1993 multipliers	0.520, 0.265	Table A2
Removals 1992, 1993	40,956, 11,392	Table 1 (2025)
Removals 1994, 1995	1,314, 413	Reconstruction
Survey index 1989, 1994	2,127,417, 21,797	Survey T3
Covariate 1990, 1991	5783, 138	Capelin file
Outcome years 2016–2021	340–440 kt	DB extract
Breach of 276 kt	by 1993 (realized)	A2 + xteNCAM

0.7 The covariate alarm and the outcome years

Proposition 8 (the covariate alarm leads the mortality crossing). *In the observed capelin series, the acoustic estimate falls from 5783 kt (1990) to 138 kt (1991) — a certified factor of $\frac{138}{5783} \approx 0.024$ — one step before the Table A2 mean natural mortality diagnostic crosses $M = 1$ (0.403 in 1990 to 1.002 in 1991, diagnostic column, not a model input). The one-year lead makes the covariate the certified probe datum of the regime shift: the monitoring instrument moved before the mortality regime did.*

Proof. Both readings are observed rows of the locked capelin file (no interpolation, per the archival discipline); the factor is their exact ratio. The mortality crossing is the diagnostic column’s consecutive values straddling 1. The proposition asserts only the temporal order of the two certified events, which the rows display directly. $\square$

Proposition 9 (the outcome years: partial recovery, no return to the reference scale). *The per-year database extension reads the stock at 340, 433, 394, 419, 440, 411 kt for 2016–2021 (same assessment family, 2021 vintage): the stock re-crosses the 276 kt auxiliary threshold by 2016 and stays above it through 2021, while its maximum (440 kt, 2020) remains at $\frac{44000}{88460} \approx 49.7\%$ of the reference point. The recovery proceeded under a modest fishery (4,436 t in 2015), completing the bidirectional reading: the dynamics, not the catch level, set the trajectory on the way up as on the way down.*

Proof. The seven values are the locked database rows for 2016–2021 (exact integers in kt); the crossing and maximum comparisons are direct rational comparisons against 276 and 884.6; the ratio $440/884.6 = 44000/88460$ is reduced exactly in the script. The 2015 removals row is the cross-checked file value of Proposition 5. $\square$

Remark 10 (the monitoring consequence). Read together: catch level is not the certifiable instrument at the realized rates (Proposition 2), while both independent monitored series — the survey index (Proposition 4) and the covariate (Proposition 8) — move before or with the collapse, and the reference machinery itself was reconstructed around a monitoring indicator after 2016 (a three-year review cycle with an interim survey indicator, the indicator value available by early January). The viable instrument is the monitoring schedule: the same review-timing reading the companion paper certifies for the 2022–2023 observation suspension (Abaee, 2026c, Section 3.10).

0.8 Delimitations

Every claim is a certificate about the realized windows of the certified series — not a forecast, not a stochastic model, and not a counterfactual catch analysis: the net multipliers conflate the mortality regime with all removals (the archival diagnostics rule), so no statement here identifies what a different catch path would have done. The biomass series are assessment outputs (model-based), treated as the data of record with their published uncertainty untouched; the survey index and the covariate are independent instruments and carry their own sampling error, untouched here. The covariate series is observed-years only; the one-year lead is a statement about the certified rows, robust to no claim about unobserved years. Single stock; two windows; no averaging beyond the reference point’s own construction.

0.9 Verification methods

Every number, fraction, inequality, power, ratio, and temporal claim in this paper is regenerated by the accompanying script in exact integer and rational arithmetic, standard library only, directly from the locked files: the six good-window and two bad-window multipliers; the removals rows of both removals files and their 2015 cross-file equality; the survey and covariate factors; the breach comparisons by integer cross-multiplication; the sixteenth and fifteenth powers of the decade’s worst multiplier; the seven reference-point ratios with their exact minimum and maximum and the mean rule; the outcome-year comparisons; and every headline number of this text asserted present verbatim.

0.10 Conclusion

Applied to the public record of the Northern cod, the calculus certifies what the narrative record suggests and the data now proves exactly: the healthy decade was already marginal at the auxiliary threshold (the reference point is its own mean); the collapse proceeded with the fishery closing (the moratorium obstruction, witnessed at two orders of magnitude of removals); the breach timeline is exactly two years from the last pre-collapse reading; two model-independent instruments — the survey and the covariate — moved before or with the regime crossing; and the partial recovery re-crossed the auxiliary threshold without approaching the reference scale. Viability on this stock was never a catch-optimization problem; it was a monitoring problem. That is the applied content of the obstruction calculus, now certified.

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Declarations

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Competing interests

None.

Data availability

All inputs are the archived public files of the calibration record (wave-e-cod extracts and the record’s CSV files), with primary sources cited at archival time.

Code availability

The verification script `applied_regime_viability_v2_verification.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it regenerates every fraction, inequality, power, ratio, and temporal claim verbatim from the locked files.

AI declaration

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